

Numbers and Operations

A short sample textbook for grade 7

Chapter 1: Integers

Integers are the whole numbers together with their opposites. They let us describe temperatures below zero, debts and depths under the sea level.

1.1 Opposite numbers

Every integer has an opposite at the same distance from zero on the number line.

- The opposite of 3 is -3 .
- The opposite of -7 is 7.
- Zero is its own opposite.

$$3 + (-3) = 0$$

The sum of a number and its opposite is always zero.

1.2 The number line

We draw the integers on a line, with the negative numbers to the left of zero and the positive numbers to the right.



Figure 1: The number line from -5 to 5 .

A number is greater than every number to its left.

Chapter 2: Fractions

A fraction names a part of a whole. The denominator tells how many equal parts the whole is cut into, and the numerator tells how many of them we take.

1. Find a common denominator.
2. Add the numerators.
3. Simplify the result.

$$1/2 + 1/4 = 3/4$$

2.1 Equivalent fractions

Two fractions are equivalent when they name the same part of the whole, like one half and two quarters.

Chapter 3: Decimals

Decimals are another way to write fractions whose denominator is a power of ten.

$$0.25 = 25/100 = 1/4$$

To compare two decimals, compare their digits from left to right.